

System Architecture Specification

Multi-Hospital Post-Discharge Clinical Outreach Platform (PRD v2.0)

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1. Executive Summary & Problem Space

Preventable 30-day hospital readmissions impose substantial clinical and financial burdens across healthcare systems. While post-discharge telephone follow-ups within 48 to 72 hours significantly lower readmission rates, clinical care teams face high call volumes, volatile staffing, and strict hospital telephony concurrency limits. Generic conversational bots are clinically unacceptable due to hallucination risks, inability to triage multi-symptom urgency, and lack of strict hospital isolation.

This platform delivers a resilient, multi-tenant clinical operations engine unifying deterministic capacity-bounded queue scheduling, clinical protocol grounding via RAG, multi-agent AI voice triage with dual independent consensus validation, a human-in-the-loop escalation desk, and real-time electronic health record (EHR) synchronization.

Architectural Invariant — Zero Silent Failures: Under no circumstance may an acute, life-threatening red-flag symptom (such as substernal chest pressure, severe dyspnea, or critical medication omission) be dropped or silenced by automated heuristics. The platform combines deterministic clinical rule validators with generative protocol reasoning via an invariant Conservative Arbiter.

2. Layer Topology & System Components

The system is architected as a modular monolith with clear domain boundaries, avoiding distributed transaction failures while supporting rapid horizontal migration to PostgreSQL and distributed worker pools:

Architectural Layer	Technologies & Frameworks	Core Responsibilities & Guarantees
Presentation Layer	Next.js 14+ App Router, React, Tailwind CSS, Lucide Icons	Real-time operational command center, multi-persona switcher (Platform Admin, Hospital Admin, Campaign Manager, Clinical Reviewer), telemetry monitors.
API & Gateway Layer	FastAPI, Pydantic V2, Python 3.13, Starlette Middleware	RESTful endpoints, JWT bearer auth, RBAC route guards, correlation ID logging, request validation, and automatic TenantContext injection.
Outbound Queue Engine	Capacity Governor, Lease Reaper, Math Priority Engine	Strict concurrency limits (max N concurrent active lines), continuous priority scoring, non-linear deadline pressure, starvation aging, idempotent claim locking.
AI Safety & Triage	Dual-Agent Pipeline (Gemini Flash / OpenAI), Consensus Arbiter	Assessment A (Protocol Reasoning LLM) + Assessment B (Deterministic Rule Validator). Conservative consensus arbitration, prompt injection defense.
Clinical Knowledge (RAG)	Vector Cosine Index, Tenant-Filtered Protocol Store	Stores verified medical literature (ACC/AHA HF, Surgical Wound, COPD). Strictly queries WHERE hospital_id = :tenant_id with chunk citations.
Persistence Layer	SQLite in WAL Mode / PostgreSQL, SQLAlchemy ORM	Write-Ahead Logging (WAL) for non-blocking concurrent reads, foreign key constraints, atomic state transitions, append-only audit event log.
EHR Integration Layer	MockEHRService, FHIR Protocol Adapter	Pluggable interface for progress note documentation and clinical observation recording. Simulates retry recovery and network latency.

3. Multi-Hospital Tenancy & Security Isolation

Healthcare multi-tenancy requires strict computational and data boundaries to prevent cross-hospital data leakage while enabling consolidated platform administration:

- **Server-Side Tenant Context:** Every inbound HTTP request resolves the authenticated user's `hospital_id` via the `TenantContext` dependency. All repository database queries automatically inject `filter(Entity.hospital_id == tenant_ctx.hospital_id)`.
- **Four-Tier Role-Based Access Control (RBAC):**
 - **PLATFORM_ADMIN:** Global observability, multi-hospital capacity governance, and tenant provisioning.
 - **HOSPITAL_ADMIN:** Hospital-scoped queue configuration, operating hours (08:00 - 20:00), and compliance audit logs.
 - **CAMPAIGN_MANAGER:** Cohort eligibility rule authoring, campaign start/pause toggling, and workload forecasting.
 - **CLINICAL_REVIEWER:** Human-in-the-loop escalation desk, dual assessment audit, verbatim quote review, and EHR resolution.
- **Hardware Concurrency Partitioning:** Telephony channel caps (e.g., 10 lines for St. Jude Hospital, 15 lines for Metro General Hospital) are tracked independently. A surge in one hospital's patient discharges cannot exhaust another hospital's calling capacity.

4. Mathematical Queue Prioritization & Concurrency Engine

Outbound queue dispatching relies on a continuous priority scoring algorithm combining clinical risk, non-linear deadline urgency, callback commitments, retry backoff, and starvation aging:

Continuous Priority Scoring Formula:

$$P(t) = w_r \cdot S_{risk} + w_d \cdot [\max(0, 1 - T_{remain} / T_{window})]^{\gamma} + w_c \cdot S_{callback} + w_b \cdot S_{retry} + \alpha \cdot T_{wait}$$

Where:

- **S_risk:** Base clinical risk tier (URGENT = 1.0, HIGH = 0.8, MEDIUM = 0.5, ROUTINE = 0.25).
- **T_remain / T_window:** Time remaining before the post-discharge clinical deadline (e.g., 48 hours).
- **γ (gamma = 2.0):** Non-linear urgency exponent. As the deadline approaches, deadline urgency rapidly eclipses baseline risk.
- **w_c · S_callback:** Priority boost for patient-requested scheduled callbacks.
- **α · T_wait (Aging Factor):** Accumulates continuously for waiting tasks, mathematically guaranteeing zero queue starvation.

- **Concurrency Cap Enforcement:** When queue dispatchers claim work, an atomic transaction counts active `CALLING` tasks for that hospital. If `active_count >= hospital.max_concurrent_calls`, scheduling halts cleanly.
- **Stale Worker Reaper & Lease Heartbeats:** Tasks in `CALLING` carry a `lease_expires_at` timestamp. If a worker process crashes mid-call, the background reaper detects the expired lease, resets the task to `PENDING`, and restores line capacity.

5. Telephony Lifecycle & State Machine

Outbound tasks navigate a strictly governed state machine rejecting invalid state transitions:

- **PENDING \$to\$ SCHEDULED \$to\$ CALLING:** Initiated by capacity-aware dispatcher.
- **CALLING \$to\$ COMPLETED:** Call connected, patient surveyed, triage completed.
- **CALLING \$to\$ RETRY_SCHEDULED:** Unanswered, busy, or dropped calls apply exponential backoff (e.g. +30m, +2h, +6h) clamped strictly within hospital calling hours (08:00 - 20:00).
- **CALLING \$to\$ ESCALATED:** Red flag or assessment disagreement triggers immediate escalation.

6. Audit Logging & HIPAA Regulatory Compliance

Every queue transition, AI decision turn, human clinical review action, and campaign status toggle generates an append-only, immutable record in the audit_events table:

- **Cryptographic Traceability:** Logs store timestamp, actor ID, action name, target entity, old state snapshot, new state snapshot, correlation ID, and clinician justification notes.
- **Zero Direct SQL for Agents:** AI models possess zero direct database execution privileges. All state modifications must pass through audited, authenticated Pydantic tool endpoints.

7. Test Verification & Safety Validation

The complete architecture is validated by automated test suites ensuring zero regression:

- **Unit & Integration Suite:** 13 of 13 tests passing (100%) in tests/ spanning RBAC, concurrency limits, starvation prevention, EHR idempotency, and tenant isolation.
- **Clinical Safety Evaluation Suite:** 30 of 30 clinical benchmark cases passing with **0.00% False Negative Rate** and 100% prompt injection block rate.